A sharp refresh scale for power-free interval counts

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Abstract

The random-sieve model of power-free integers assigns an independent uniform forbidden residue class modulo p^k to each prime p. We determine how the centered count in an interval of length H responds when each prime coordinate is independently resampled with probability ε_H. For every fixed integer $k \geq 2$, the correlation between the original and resampled counts tends to e^{-t} when $\varepsilon_H \log\log H \to t \in [0, \infty)$, and tends to zero when this product diverges. Thus $1/\log\log H$ is the sharp refresh scale. The proof expresses the covariance as a positive arithmetic sum and transfers the classical normal order of the number of distinct prime factors to its bounded weights. An exact cancellation of Euler factors then gives the correlation profile. As a consequence, the number of prime coordinates in a character sampled according to the interval count's normalized Fourier energy, divided by $\log\log H$, converges in probability to one.

1 The model and the refresh question

A power-free sieve excludes one residue class modulo p^k for each prime p. For the usual k-free integers that class is zero; in the stationary random-sieve model, the forbidden classes are independent and uniform. Grimmett [1] formulated this model for pairwise coprime sieving moduli and related it to the distribution of arithmetic translates. For squarefree integers, Cellarosi and Sinai [2] identified the associated dynamical system with a translation on a compact product group. These descriptions retain the separate contribution of each prime, so they also allow us to perturb those contributions independently.

The fluctuations of interval counts in this model have an established static theory. Their variance has order $H^{1/k}$, and the results of Gorodetsky, Mangerel and Rodgers [3] give variance asymptotics, Gaussian limits and spatial functional limits in a more general B-free setting. Here we ask how the same interval statistic changes when some of its prime-residue coordinates are resampled. The interval remains fixed while the forbidden residue classes change; the two resulting counts therefore have the same marginal distribution, but their dependence decreases as more coordinates are refreshed.

Fix an integer $k \geq 2$. Let

$$\mathcal{G}_k = \prod_{p \text{ prime}} \mathbb{Z}/p^k\mathbb{Z},$$

with its product uniform probability measure, and write $U = (U_p)_p$ for a random element. For $n \in \mathbb{Z}$, define

$$\eta_n(U) = \prod_p \left(1 - \mathbf{1}_{\{U_p \equiv n \pmod{p^k}\}}\right), \qquad \rho_k = \prod_p (1 - p^{-k}) = \zeta(k)^{-1}. \tag{1.1}$$

The infinite product defining η_n is the decreasing limit of finite products of indicators. Since $\sum_p p^{-k} < \infty$, its expectation is $\rho_k > 0$. For an integer $H \geq 1$, our statistic is the centered

count

$$S_H(U) = \sum_{n=1}^{H} (\eta_n(U) - \rho_k). \tag{1.2}$$

Given $\varepsilon \in [0,1]$, construct U^ε by retaining U_p with probability $1-\varepsilon$ and otherwise drawing a fresh uniform residue modulo p^k, independently for each prime. The fresh draw is also independent of U, and may equal the old residue. This is the usual independent-coordinate noise operation on a product probability space [4, §8.2]. It preserves the law of U and acts on entire residue classes, as the two-prime illustration in Figure 1 shows.

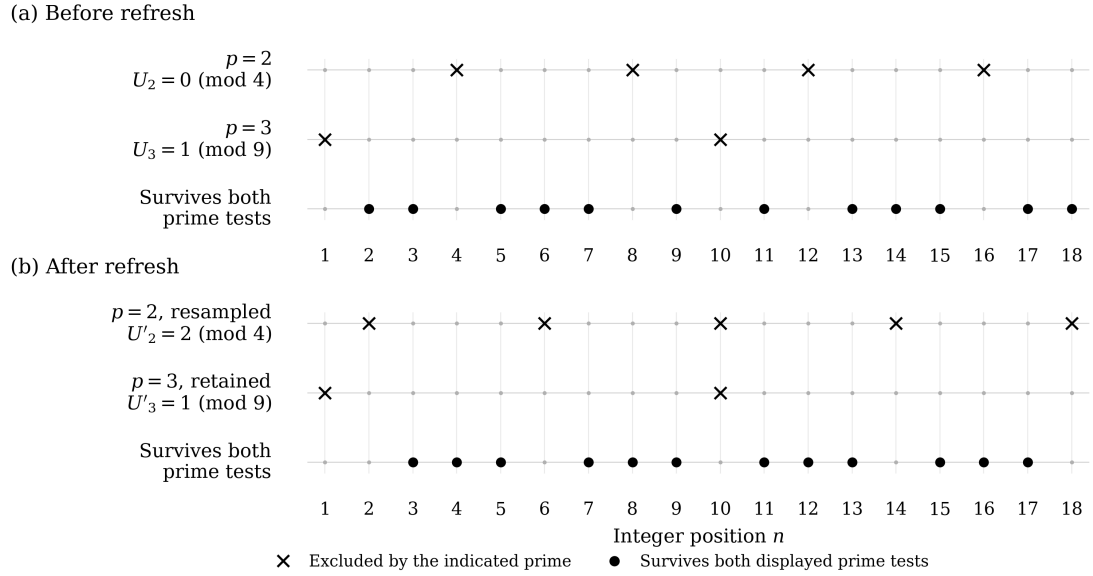


Figure 1: A finite illustration with $k = 2$ and primes $2, 3$: the forbidden class modulo 4 is resampled from 0 to 2, while the class 1 modulo 9 is retained. Each prime row marks the sites excluded by that coordinate, and the survivor row combines the two exclusions. The theorem uses all primes; this diagram illustrates the update rule only.

Define the normalized covariance

$$R_{H,k}(\varepsilon) = \frac{\mathbb{E}[S_H(U)S_H(U^\varepsilon)]}{\mathrm{Var}(S_H(U))}. \tag{1.3}$$

The variance is positive for every $H \geq 1$, as will follow from the positive sum in Section 2. Because the two marginal laws agree, $R_{H,k}$ is their correlation coefficient.

Theorem 1.1 (sharp refresh scale)**.** *Let $k \geq 2$ be fixed, let $H \to \infty$ through the positive integers, and let $\varepsilon_H \in [0,1]$. If*

$$\varepsilon_H \log\log H \longrightarrow t \in [0, \infty),$$

then

$$R_{H,k}(\varepsilon_H) \longrightarrow e^{-t}. \tag{1.4}$$

If $\varepsilon_H \log\log H \to \infty$, then $R_{H,k}(\varepsilon_H) \to 0$.

In particular, a refresh probability of smaller order than $1/\log\log H$ preserves asymptotically all correlation, while a probability of larger order than that scale gives decorrelation. At the scale itself, the parameter t determines the limiting profile. All assertions concern fixed k and the product probability model just defined.

The addition to the static theory is the dependence on the refresh parameter as H grows. The variance formula of [3, Lemma 3.4] supplies the relevant arithmetic structure at zero refresh, and the general product-noise formula [4, Proposition 8.28] describes how a character is damped according to its coordinate-support size. To obtain (1.4), one must determine how the interval count's energy is distributed among those sizes. We do this through a positive covariance expansion: its arithmetic weights concentrate on integers of size $H^{1/k}$, and the Hardy–Ramanujan normal-order theorem [5] transfers to these weights. The resulting prime-factor count is asymptotic to $\log\log H$ in probability, which produces the exponential profile. Section 5 states the corresponding concentration result directly in terms of Fourier energy.

2 A positive formula for the covariance

We first keep track of the refresh operation at one prime. Write $z = 1 - \varepsilon$ for the retention probability and, for $q = p^k$, consider two sites n, m. If the coordinate is retained, both sites survive with probability $1 - 2/q + q^{-1}\mathbf{1}_{\{q|m-n\}}$. If it is refreshed, the two relevant residues are independent and the probability is $(1 - 1/q)^2$. Their mixture is therefore

$$1 - \frac{2}{q} + \frac{1-z}{q^2} + \frac{z}{q}\mathbf{1}_{\{q|m-n\}}. \tag{2.1}$$

To separate its constant and divisibility parts, put

$$A_p(z) = 1 - 2p^{-k} + (1-z)p^{-2k}, \qquad A_k(z) = \prod_p A_p(z),$$

and define the squarefree-supported weight

$$f_z(d) = \mu^2(d) \prod_{p|d} \frac{z}{A_p(z)}, \qquad f_z(1) = 1. \tag{2.2}$$

Here μ is the Möbius function, so $\mu^2(d)$ is the indicator of squarefree integers. Throughout the proof, constants implicit in $\ll_k$ and $\asymp_k$ may depend on fixed k, but not on H or $z \in [0,1]$.

Since $A_p(z) \geq 1 - 2p^{-k} \geq 1/2$, the product $A_k(z)$ is positive. Moreover,

$$0 \leq f_z(d) \leq D_k \mu^2(d), \qquad D_k = \prod_p (1 - 2p^{-k})^{-1} < \infty. \tag{2.3}$$

The summability of p^{-k} gives convergence of these products. Expanding the product of the local probabilities (2.1) now yields

$$\mathbb{E}[\eta_n(U)\eta_m(U^\varepsilon)] = A_k(z) \sum_{d \geq 1} \frac{f_z(d)}{d^k} \mathbf{1}_{\{d^k|m-n\}}. \tag{2.4}$$

The series is absolutely convergent, uniformly in z, by (2.3) and $k \geq 2$. One may first establish (2.4) for finitely many primes and then pass to the limit; the indicator products converge in L^2, and the displayed bound controls the series.

The same expansion also identifies the term that centering removes. The local identity

$$A_p(z) + zp^{-2k} = (1 - p^{-k})^2$$

gives

$$\rho_k^2 = A_k(z) \sum_{d \geq 1} \frac{f_z(d)}{d^{2k}}. \tag{2.5}$$

After subtracting (2.5) from (2.4), the sum over the two sites reduces to an elementary residue count. For an integer $q \geq 1$, write $H = aq + r$, where $0 \leq r < q$. Exactly r residue classes have $a + 1$ representatives in $\{1, \ldots, H\}$ and the other $q - r$ have a. Consequently,

$$\begin{aligned} \sum_{1 \leq n,m \leq H} \mathbf{1}_{\{q \mid m-n\}} - \frac{H^2}{q} &= r(a+1)^2 + (q-r)a^2 - \frac{H^2}{q} \\ &= q\psi(H/q), \end{aligned} \tag{2.6}$$

where

$$\psi(x) = \{x\}(1 - \{x\})$$

and $\{x\}$ denotes the fractional part. Combining (2.4)–(2.6) gives the required positive formula.

Proposition 2.1. *For $H \geq 1$ an integer and $z \in [0, 1]$,*

$$C_{H,k}(z) := \mathbb{E}[S_H(U) S_H(U^{1-z})] = A_k(z) \sum_{d \geq 1} f_z(d) \psi(H/d^k). \tag{2.7}$$

The series converges absolutely. Indeed, for $d > H^{1/k}$ its kernel is at most H/d^k, and (2.3) applies. At $z = 1$, every prime $d > H^{1/k}$ gives a strictly positive summand, proving the asserted positivity of the variance. At $z = 0$, only $d = 1$ can contribute, and its kernel is $\psi(H) = 0$, in agreement with independence after complete refresh.

Formula (2.7) adapts the established positive arithmetic variance representation to the two-copy coupling. Its role is to retain the prime-factor dependence of the noise. The next section determines where its unperturbed weight is concentrated; Section 4 will then compare the perturbed and unperturbed sums.

3 Prime-factor normal order under the variance weights

Set $T = H^{1/k}$ and $f = f_1$. At zero refresh, the sum in (2.7) is

$$W_H = \sum_{d \geq 1} f(d) \psi((T/d)^k). \tag{3.1}$$

We need its order of magnitude and enough control of its tails to transfer a density-one arithmetic statement. The full static variance asymptotic is not required.

Lemma 3.1. *As $H \to \infty$,*

$$W_H \asymp_k T.$$

If

$$\nu_H(d) = \frac{f(d)\psi((T/d)^k)}{W_H}, \tag{3.2}$$

then ν_H is a probability measure on the positive integers, and for $0 < \eta < 1 < M$,

$$\nu_H(d < \eta T) \ll_k \eta, \qquad \nu_H(d > MT) \ll_k M^{1-k}. \tag{3.3}$$

Proof. On $d \le T$, the kernel is at most $1/4$, whereas on $d > T$ it is at most $(T/d)^k$. Together with $f \le D_k$, these bounds give

$$W_H \ll_k T + T^k \sum_{d>T} d^{-k} \ll_k T.$$

For the lower bound, restrict to squarefree $d \in [3T/2, 2T]$. On this interval $f(d) \ge 1$ and

$$\psi((T/d)^k) \ge 2^{-k}(1 - (2/3)^k) > 0.$$

There are $\gg T$ such squarefree integers. To recall the elementary count, the identity $\mu^2(n) = \sum_{a^2|n} \mu(a)$ gives

$$\sum_{n \le x} \mu^2(n) = \sum_{a \le \sqrt{x}} \mu(a) \left\lfloor \frac{x}{a^2} \right\rfloor = \frac{x}{\zeta(2)} + O(\sqrt{x}).$$

Subtracting this formula at the endpoints proves the lower bound. Finally, summing the bound $D_k/4$ over $d < \eta T$, and the bound $D_k(T/d)^k$ over $d > MT$, then dividing by $W_H \gg_k T$, gives (3.3). □

Let $\omega(d)$ denote the number of distinct prime divisors of d, with $\omega(1) = 0$. For the uniform distribution on integers up to N, its normal order is $\log\log N$ [5]. The covariance weights are not uniform, but their boundedness and the normalization in Lemma 3.1 allow the same first-order statement to survive.

Lemma 3.2. *With $\lambda_H = \log\log H$,*

$$\frac{\omega(d)}{\lambda_H} \longrightarrow 1 \qquad \textit{in probability under } \nu_H. \tag{3.4}$$

Proof. We first record the form of the classical normal-order estimate needed here. For every fixed $\delta > 0$,

$$\#\left\{ n \le N : \left| \frac{\omega(n)}{\log\log N} - 1 \right| > \delta \right\} = o(N). \tag{3.5}$$

For completeness, a second-moment argument proves this from Mertens' estimate $\sum_{p \le x} p^{-1} = \log\log x + O(1)$ [6]. Put $y = N^{1/3}$,

$$L_y = \sum_{p \le y} \frac{1}{p}, \qquad \Omega_y(n) = \sum_{p \le y} \mathbf{1}_{\{p|n\}}.$$

Counting multiples gives $\sum_{n \le N} \Omega_y(n) = NL_y + O(\pi(y))$, where $\pi(y)$ is the number of primes at most y. Expanding the second moment and separating equal from distinct primes gives

$$\sum_{n \le N} \left(\Omega_y(n) - L_y\right)^2 \ll NL_y + \pi(y)^2 + L_y\pi(y) \ll N\log\log N. \tag{3.6}$$

Here each distinct-prime count is $\lfloor N/(pq) \rfloor$, so the total floor error is $O(\pi(y)^2)$; the elementary bound $\pi(y) \le y$ suffices for the last inequality. An integer at most N has at most two distinct prime factors larger than $N^{1/3}$. Thus $|\omega(n) - \Omega_y(n)| \le 2$, while $L_y = \log\log N + O(1)$. Chebyshev's inequality proves (3.5), with exceptional-set proportion $O_\delta(1/\log\log N)$.

Now fix $0 < \eta < 1 < M$. Since k is fixed,

$$\log\log(MT) = \lambda_H - \log k + o(1).$$

Applying (3.5) with $N = \lfloor MT \rfloor$ shows that only $o(T)$ integers in $[\eta T, MT]$ have $|\omega(d)/\lambda_H - 1| > \delta$. Each has ν_H-mass at most $D_k/(4W_H) \ll_k T^{-1}$, so their total mass tends to zero. By (3.3),

$$\limsup_{H\to\infty} \nu_H\left(\left|\frac{\omega(d)}{\lambda_H} - 1\right| > \delta\right) \ll_k \eta + M^{1-k}.$$

Letting $\eta \downarrow 0$ and $M \uparrow \infty$ proves (3.4). □

This transfer does not require independence between squarefreeness, the oscillating kernel and $\omega(d)$. The exceptional integers remain negligible because their number is $o(T)$, their individual weights are bounded, and the total weight has order T.

4 From the covariance weights to the refresh profile

We can now compare the noisy sum with its variance normalization. Define

$$c_p = \frac{1}{p^{2k}(1-2p^{-k})}, \qquad B_k = \sum_p c_p < \infty.$$

For squarefree d, cancellation in (2.2) gives the exact identity

$$\frac{A_k(z) f_z(d)}{A_k(1) f(d)} = z^{\omega(d)} \prod_{p\mid d}(1 + \varepsilon c_p), \qquad z = 1 - \varepsilon. \tag{4.1}$$

Indeed, a prime dividing d contributes the factor z after its $A_p(z)$ terms cancel, while a prime not dividing d contributes $A_p(z)/A_p(1) = 1 + \varepsilon c_p$. The remaining product lies between 1 and $e^{\varepsilon B_k}$, uniformly in d. Dividing (2.7) by its value at $z = 1$ therefore yields

$$\mathbb{E}_{\nu_H}\big[(1-\varepsilon)^{\omega(d)}\big] \le R_{H,k}(\varepsilon) \le e^{\varepsilon B_k}\mathbb{E}_{\nu_H}\big[(1-\varepsilon)^{\omega(d)}\big]. \tag{4.2}$$

The measure ν_H is supported on squarefree integers, so (4.1) applies to every nonzero summand. At $d = 1$ we use the empty-product convention $z^0 = 1$, including $z = 0$; its weight is zero because H is an integer.

Proof of Theorem 1.1. Suppose first that $\varepsilon_H \lambda_H \to t < \infty$. Then $\varepsilon_H \to 0$ and

$$s_H = -\lambda_H \log(1 - \varepsilon_H) \longrightarrow t,$$

because $-\log(1-u) = u + O(u^2)$ near zero. The integrand in (4.2) is

$$\exp\left(-s_H \frac{\omega(d)}{\lambda_H}\right),$$

which converges in ν_H-probability to e^{-t} by Lemma 3.2. These functions take values in $[0,1]$, so their expectations also converge to e^{-t}: split the expectation into any fixed deviation event and its complement, then let the deviation tolerance tend to zero. The correction factor $e^{\varepsilon_H B_k}$ tends to one, and (4.2) proves (1.4), including $t = 0$.

If instead $\varepsilon_H \lambda_H \to \infty$, split the same expectation according to whether $\omega(d) < \lambda_H/2$. On the complementary event,

$$(1-\varepsilon_H)^{\omega(d)} \le e^{-\varepsilon_H \lambda_H/2}.$$

The upper bound in (4.2) consequently gives

$$0 \le R_{H,k}(\varepsilon_H) \le e^{B_k}\left[\nu_H\big(\omega(d) < \lambda_H/2\big) + e^{-\varepsilon_H \lambda_H/2}\right] \longrightarrow 0.$$

The first term vanishes by Lemma 3.2 and the second by the assumed growth of $\varepsilon_H \lambda_H$. This argument includes complete refresh, $\varepsilon_H = 1$, and proves the remaining regime. □

5 Concentration of Fourier support

The refresh profile also identifies the number of prime coordinates carrying the interval statistic's Fourier energy. This gives a spectral interpretation of the scale in Theorem 1.1 without introducing a different perturbation model.

Let $\widehat{\mathcal{G}_k}$ be the characters of the compact product group. Each character is nonconstant on only finitely many prime coordinates; write $j(\chi)$ for their number. These characters form an orthonormal basis of $L^2(\mathcal{G}_k)$: cylinder functions are dense, and each finite coordinate product has its usual finite Fourier basis. Write

$$\widehat{S}_H(\chi) = \mathbb{E}[S_H(U)\overline{\chi(U)}].$$

Since S_H is centered, its constant coefficient is zero. Parseval's identity therefore makes

$$\mathbb{P}(J_H = j) = \frac{1}{\mathrm{Var}(S_H)} \sum_{\chi:\, j(\chi)=j} |\widehat{S}_H(\chi)|^2, \qquad j \ge 1, \tag{5.1}$$

a probability distribution. The random variable J_H describes a character sampled by its normalized Fourier energy; it does not count the coordinates actually refreshed in a sample of U^{ε}.

Corollary 5.1. *For each fixed $k \ge 2$,*

$$\frac{J_H}{\log\log H} \longrightarrow 1 \qquad \textit{in probability.} \tag{5.2}$$

Proof. Conditional expectation under independent refresh multiplies a character by $(1-\varepsilon)^{j(\chi)}$: a fresh uniform coordinate has zero expectation in every nontrivial character, while a retained coordinate contributes its old value. The product-noise identity [4, Proposition 8.28], extended from finite products by L^2 approximation, gives

$$\begin{aligned} R_{H,k}(\varepsilon) &= \sum_{\chi \ne 1} \frac{|\widehat{S}_H(\chi)|^2}{\mathrm{Var}(S_H)} (1-\varepsilon)^{j(\chi)} \\ &= \mathbb{E}[(1-\varepsilon)^{J_H}]. \end{aligned} \tag{5.3}$$

For fixed $t \ge 0$, set $\varepsilon_H = 1 - e^{-t/\lambda_H}$. Then $\varepsilon_H \lambda_H \to t$, so Theorem 1.1 implies

$$\mathbb{E}[e^{-tJ_H/\lambda_H}] \longrightarrow e^{-t}.$$

Using this at $t = 1$ and $t = 2$ gives

$$\mathbb{E}\left[\left(e^{-J_H/\lambda_H} - e^{-1}\right)^2\right] \longrightarrow 0.$$

For every fixed $\delta > 0$, the function $x \mapsto e^{-x}$ separates 1 by a positive amount from the set $\{x \ge 0 : |x-1| \ge \delta\}$. Markov's inequality now proves (5.2). □

The auxiliary distribution ν_H and the Fourier-energy distribution in (5.1) need not coincide term by term. The covariance comparison transfers the arithmetic normal order to the refresh profile, and (5.3) then transfers that profile to spectral-support concentration. This sequence is sufficient for the two-copy correlation theorem; no assertion about a Gaussian process indexed by refresh time is needed.

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